PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 2 · Probability, the binomial, and confidence intervals

TUESDAY SEP 1 AND THURSDAY SEP 3, 2026

OBJECTIVES

WHAT YOU WILL BE ABLE TO DO BY TUESDAY NIGHT

1. Identify scenarios in which a conditioned probability is necessary and calculate it.
2. Use the binomial distribution to calculate the chance that a shooter makes a given number of free throws.
3. Build a 95% confidence interval on a batting average, and say what it means.
4. Decide whether two rates are far enough apart to treat as different.
5. Adjust a kicker's make rate for the difficulty of the kicks behind it.

SETUP

UPDATE BOTH FOLDERS BEFORE WE START

1. **Open Claude Code in your repo folder, spax402-your-username.**

   Your spax402 folder holds two folders. This one is your personal repository, where you commit your work. Next to it is spax402-course-materials, our shared materials, which Professor Davis updates weekly.

2. **Run one prompt.**

   The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

   > Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3. **Confirm that weeks/week02/ now exists in your repo.**

   It holds this week's README and the worksheet you fill in on Thursday.

01

PROBABILITY

PROBABILITY

The share of times an event happens, out of all the times it could have happened.

Written P(event). Always between 0 and 1, and the probabilities of every possible outcome add to 1.

.925

Connor Hellebuyck stopped 92.5% of the 1,664 shots he faced in 2024-25.

This is a count of what already happened, not a forecast of the next shot. Read it as an estimate of how good he is. The rest of the week is about determining how good of an estimate and how to strengthen it.

NHL public stats API, 2024-25 regular season, saves by strength

THE COMPLEMENT RULE AND THE MULTIPLICATION RULE

P(save) = saves / shots faced = 1539 / 1,664 = .925

P(goal) = 1 – P(save) = .075

P(three saves in a row) = .925 x .925 x .925 = .791

Multiplying the rate by itself assumes the three shots are independent of each other and that each one carries the same chance.

? HE SAVED 92.5% OF ALL SHOTS. ON A POWER PLAY, IS THAT NUMBER HIGHER OR LOWER, AND BY HOW MUCH?

CONDITIONAL PROBABILITY

The chance of an event, restricted to the times some condition held.

Written $P(A \mid B)$, read “the probability of A given B”. Compute it by throwing away every case where B did not hold and taking the share among what is left.

THREE SAVES IN A ROW, CONDITIONED ON THE SITUATION

P(three in a row) = .925 x .925 x .925 = .791

P(three in a row | even strength) = .935 cubed = .818

P(three in a row | power play) = .855 cubed = .625

THE CONDITION MOVES THE ANSWER 19 PERCENTAGE POINTS.

WHAT ELSE WOULD YOU CONDITION ON, IF YOU WERE SCOUTING A GOALIE?

Name a condition that would better assess the quality of a goalie.

What is the cost of adding conditions?

02

ADJUSTING A RATE FOR THE ATTEMPTS BEHIND IT

LEAGUE SAVE PERCENTAGE BY SITUATION, 2024-25

SITUATION	SHOTS	SAVES	LEAGUE SAVE %	SHARE OF SHOTS
Even strength	45,837	41,759	91.1%	83.4%
Power play	7,709	6,583	85.4%	14.0%
Shorthanded	1,437	1,305	90.8%	2.6%

An even strength shot and a power play shot are not the same event. Any ranking that treats them as one is ranking how often a goalie's team took penalties, as much as it is ranking goaltending. League wide, 14.0% of shots come on the power play, and that is the share every goalie gets compared against.

NHL public stats API, 2024-25 regular season, saves by strength

ADJUSTED SAVE PERCENTAGE

Hellebuyck stopped .935 of even strength shots, .855 on the power play, .936 shorthanded.

The league faced 83.4% of its shots at even strength, 14.0% on the power play, 2.6% shorthanded.

$(0.834 \times .935) + (0.140 \times .855) + (0.026 \times .936)$

What he would have saved from a league-average mix of shots.

= .924 ADJUSTED, AGAINST .925 RAW

SAVE PERCENTAGE AND POWER PLAY SHARE FOR TWO GOALIES

GOALIE	SHOTS	POWER PLAY SHARE	SAVE %	ADJUSTED SAVE %
Adin Hill	1,293	11.7%	.906	.905
Karel Vejmelka	1,509	17.2%	.904	.906

Adin Hill has the better raw save percentage. Karel Vejmelka faced 17.2% of his shots on the power play against 11.7%, which is the widest gap in the league, and the power play is where the league saves 85.4% instead of 91.1%. On a league average mix of shots Karel Vejmelka comes out ahead.

NHL public stats API, 2024-25 regular season, saves by strength

BATTING AVERAGE AGAINST BY BATTER HAND, 2009

PITCHER	AGAINST RIGHT HANDERS	AGAINST LEFT HANDERS
Josh Beckett	.226	.258
Johan Santana	.235	.267

Beckett gave up a lower average against right handers and a lower average against left handers. Which pitcher had the lower average against overall?

Severini (2020), Analytic Methods in Sports

03

THE BINOMIAL DISTRIBUTION

BINOMIAL DISTRIBUTION

The chance of getting exactly a given number of successes, out of a fixed number of attempts, when every attempt has the same probability.

Three conditions: the number of attempts is fixed, each attempt succeeds or fails, and the probability does not change from one attempt to the next. Free throws satisfy all three about as well as anything in sport.

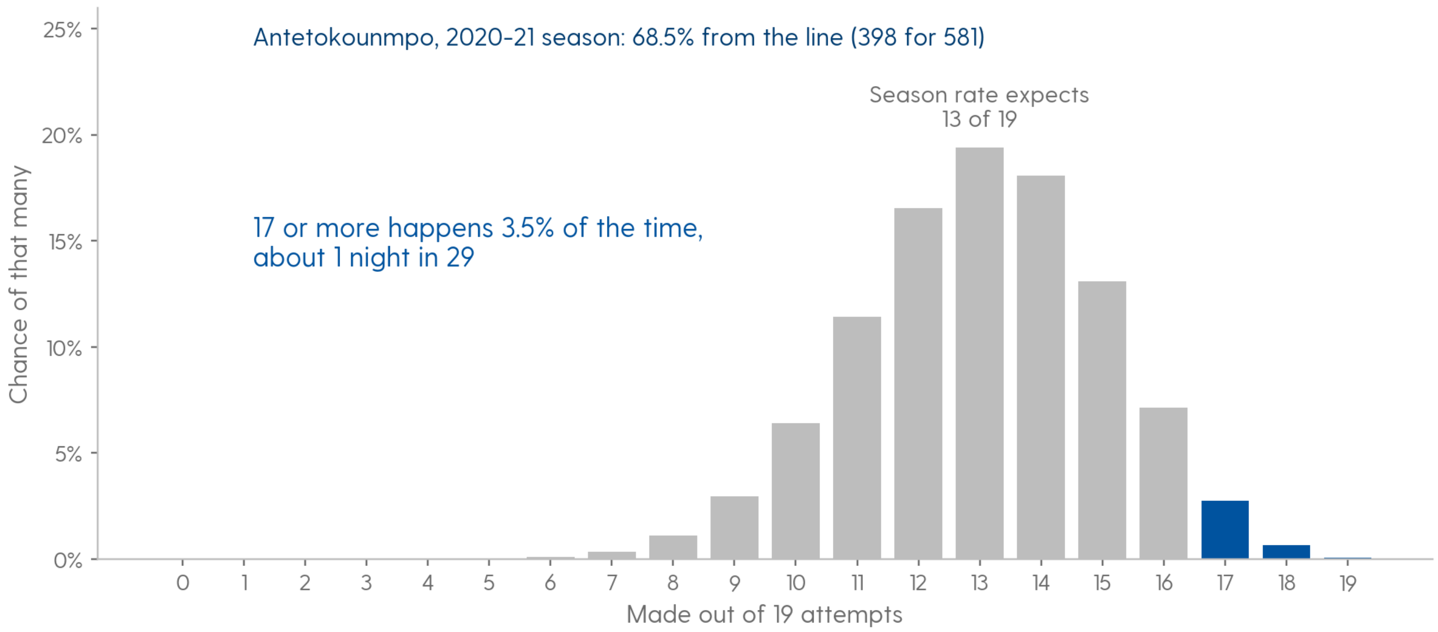
FREE THROW RATES FOR TWO SHOOTERS

SHOOTER	MADE	ATTEMPTS	RATE	95% INTERVAL
Giannis Antetokounmpo	436	707	61.7%	58.1% to 65.2%
Shai Gilgeous-Alexander	601	669	89.8%	87.5% to 92.1%

These two intervals do not overlap, so the gap between them is larger than sampling noise alone accounts for. That is all it establishes. It does not tell you the gap is large enough to matter, that it will hold next season, or that free throws are why either team wins. Most of the comparisons in this week's work will not clear even this bar.

Basketball-Reference, 2024-25 season player totals

ONE NIGHT ON THE LINE, AGAINST THE SEASON BEHIND IT

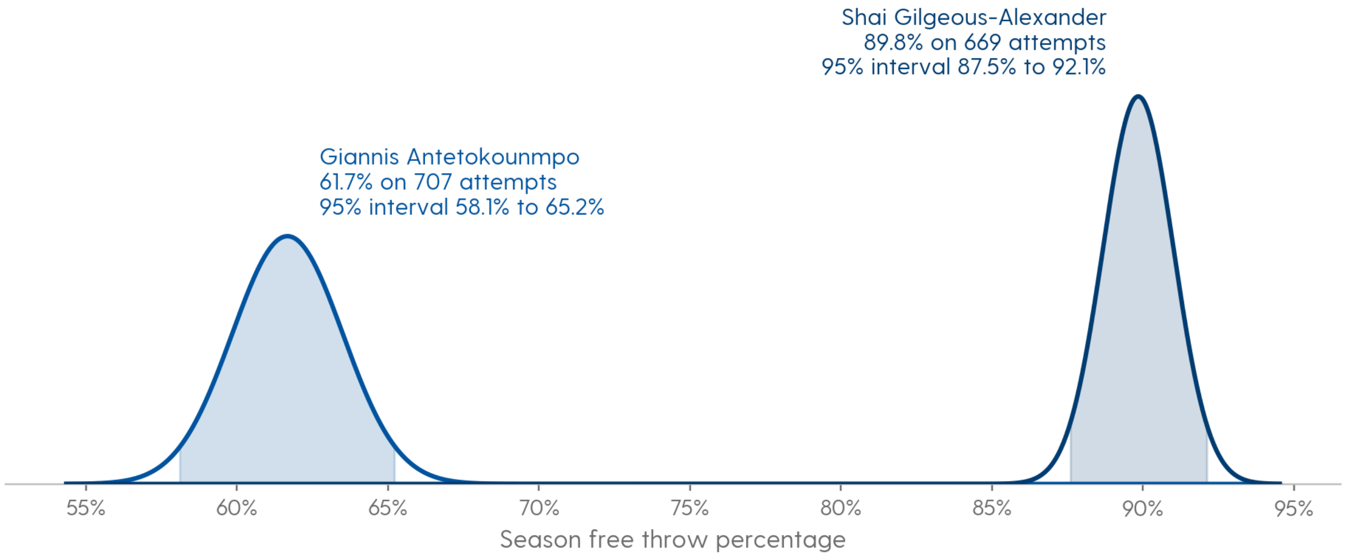


Antetokounmpo went 17 for 19 from the line in Game 6 of the 2021 NBA Finals, on his way to 50 points and 14 rebounds in the game that won the title. He shot 68.5% from the line that season, and at that rate the middle of a 19 attempt night is 13 makes. His season rate still explains the night. It sits in the right tail of this distribution, and a night that good turns up about once in 29.

Basketball-Reference, 2021 Finals Game 6 box score and 2020-21 season player totals

BINOMIAL

THE DISTRIBUTION BEHIND EACH OF THOSE TWO RATES



Each curve is where that shooter's season rate could have landed at his own number of attempts, with the shaded part covering 95% of it. Antetokounmpo took more attempts, 707 against 669, and still has the wider curve: p x (1 - p) is largest near 50%, so a rate in the middle carries more noise per attempt than one near the edge.

Basketball-Reference, 2024-25 season player totals

? GIANNIS IS AT THE LINE FOR TWO, 61.7% FROM THE LINE ALL SEASON. WHAT IS THE CHANCE HE MAKES BOTH?

BINOM.DIST IN EXCEL

1. Write =BINOM.DIST(successes, attempts, probability, FALSE) for exactly that many successes.

   =BINOM.DIST(2, 2, .617, FALSE) returns .380, which is the number you just computed by hand.

2. Choose the fourth argument deliberately.

   FALSE gives exactly that many successes. TRUE gives that many or fewer, which is a running total and rarely what a question about one specific outcome is asking for.

3. Write out the outcomes you want in words, then translate them.

   “At least 8 of 10” is 1 minus the TRUE version evaluated at 7.

04

CONFIDENCE INTERVALS

STANDARD ERROR OF A RATE

How far a rate computed from a sample typically lands from the real rate.

$$SE = \sqrt{\,p\,(1 - p) / n\,}$$

For a rate p from n attempts, which is the binomial case. More attempts give a smaller standard error, and the shrinking goes with the square root of n rather than with n itself. In Excel that is =SQRT(p*(1-p)/n), which is the formula you will write on Thursday. Divide any gap by the standard error and you get the number of standard errors it spans, which is called a z score: a rate two standard errors above the league sits at z = 2, and its 95% interval clears the league rate.

THE INTERVAL, WORKED ON A BATTING AVERAGE

p = 179 hits / 541 at bats = .331

standard error = square root of (.331 x .669 / 541) = 0.0202

margin of error = 1.96 x 0.0202 = .040

95% interval = .291 to .370

Treat 1.96 as the multiplier that turns a standard error into a 95%% interval, and memorise it.

AARON JUDGE HIT .331, SOMEWHERE BETWEEN .291 AND .370

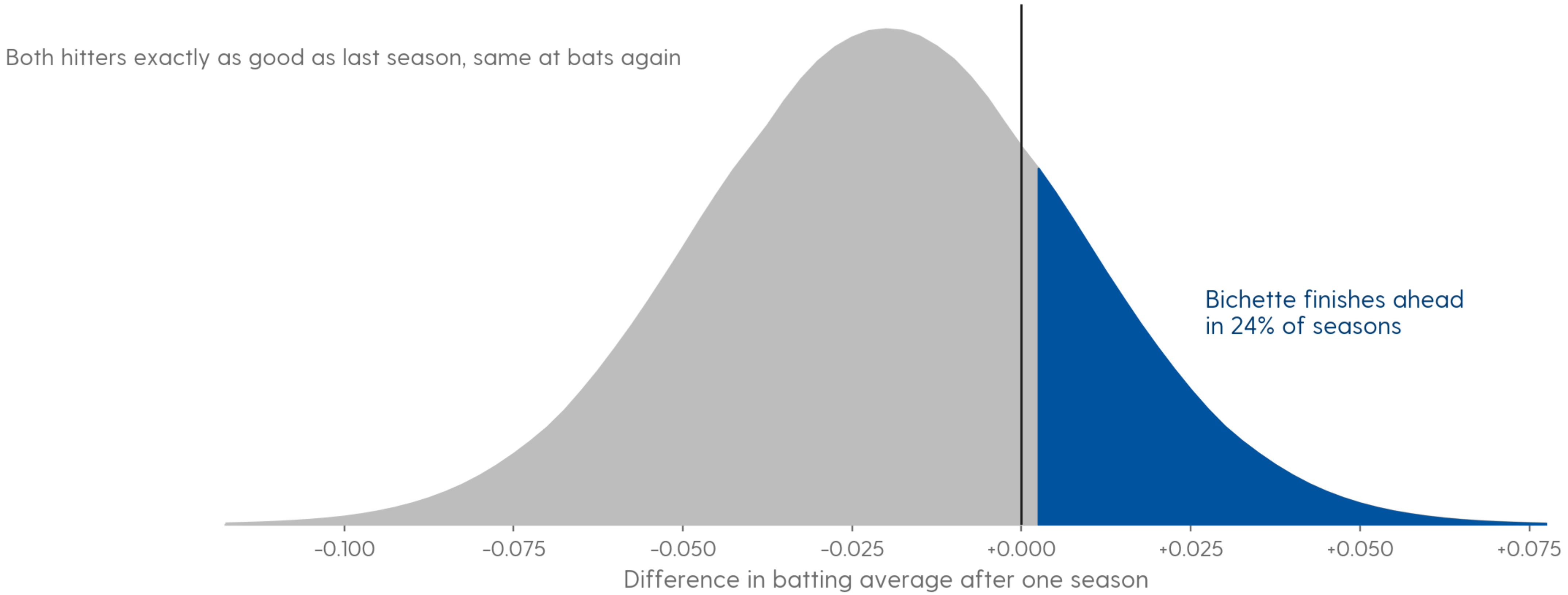
THE TOP TWO HITTERS IN BASEBALL LAST SEASON

HITTER	HITS	AT BATS	AVERAGE	95% INTERVAL
Aaron Judge	179	541	.331	.291 to .370
Bo Bichette	181	582	.311	.273 to .349

The gap between them is 2 percentage points, which baseball would call 20 points of batting average. Run both seasons again, with each hitter exactly as good as he just was, and Bichette finishes with the higher average 24% of the time.

MLB Stats API, 2025 regular season, qualified hitters

HOW OFTEN THE SECOND-PLACE HITTER FINISHES AHEAD



MLB Stats API, 2025 regular season, qualified hitters

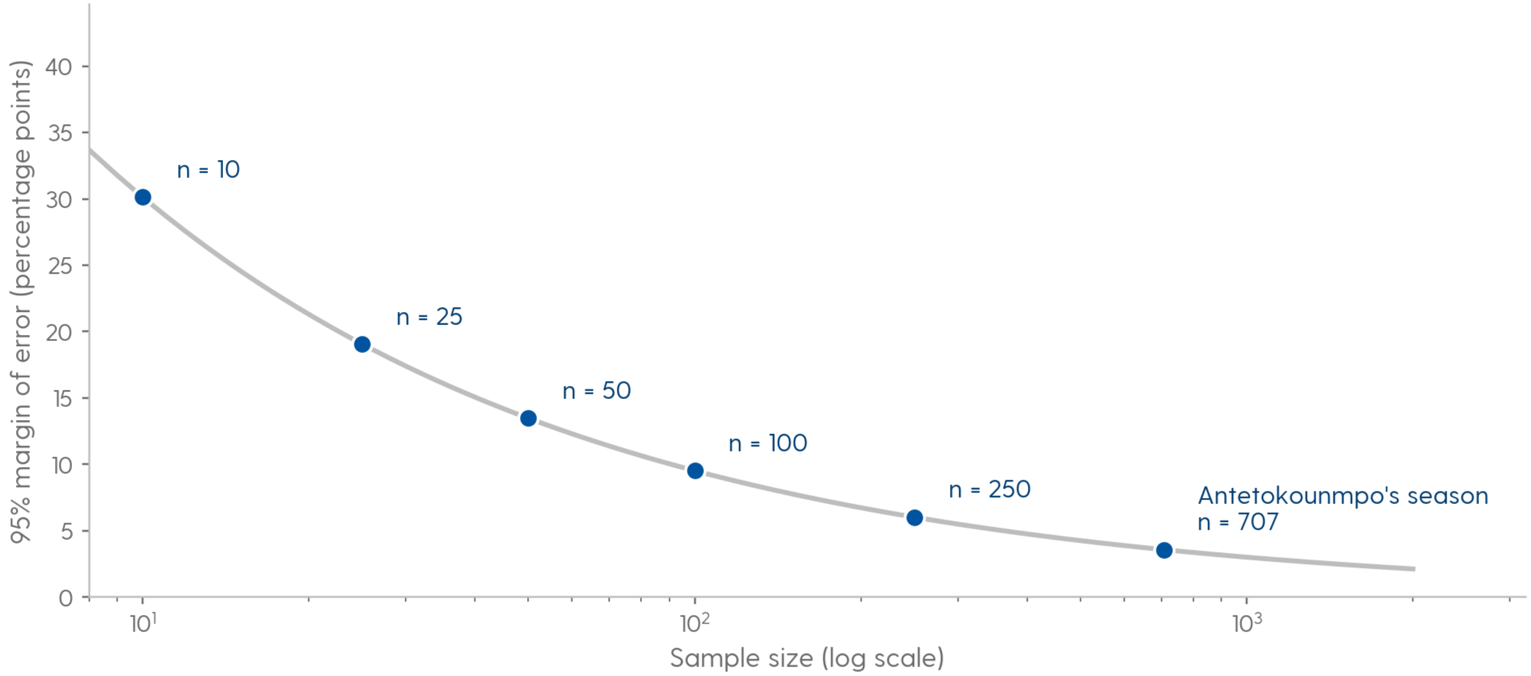
SAMPLE SIZE IS CRITICAL

Almost all analysis is constrained by sample size. This is a problem you will repeatedly encounter. You must find ways to speak with coaches about it, apply strategies to increase stability, and go beyond the data to determine significance.

CONFIDENCE INTERVAL

A range that would contain the observed rate in 95 out of 100 seasons with this sample size.

HOW THE MARGIN OF ERROR SHRINKS AS ATTEMPTS INCREASE



Every point is the same shooter at a different number of attempts, so the only thing moving is the sample. Ten free throws carry a margin of error of 30 percentage points. His actual 707 carry 3.6. The first hundred attempts buy more than the next six hundred.

Computed from the examples in this deck

YOU HAVE TO PREDICT HOW A FREE THROW SHOOTER WILL SHOOT IN THE SECOND HALF OF THE SEASON. HE HAS TAKEN 25 ATTEMPTS IN THE FIRST HALF. WHAT OTHER INFORMATION WOULD YOU WANT TO GATHER TO MAKE YOUR PREDICTION?

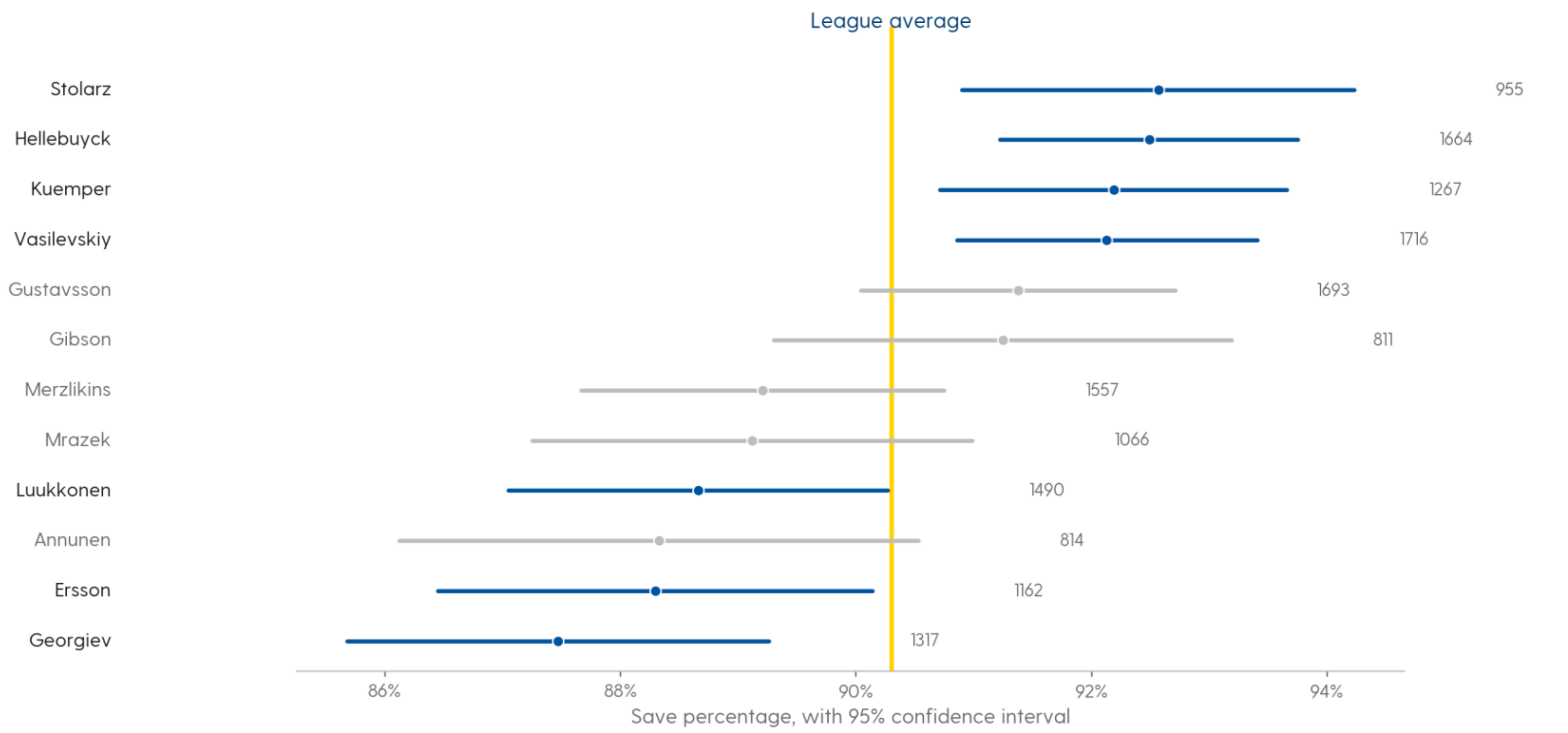
Which of the things you named happens often enough to have a usable sample?

How many attempts would you need before you stopped asking for anything else?

SAMPLE SIZE DECIDES WHAT A RANKING IS ALLOWED TO TELL YOU

A starting goalie faces around a thousand shots a season and a kicker takes about thirty field goals. Both get ranked one through thirty-two every week, and only one of those rankings has the attempts behind it to carry that much weight.

GOALIE SAVE PERCENTAGES, 800 OR MORE SHOTS EACH



Of the 42 goalies who faced at least 800 shots, 4 sit clear of league average and 3 sit clear below it. The remaining 35 have intervals that cover it.

NHL public stats API, 2024-25 regular season, saves by strength

STANDARD ERROR OF AN AVERAGE

How far an average computed from a sample typically lands from the real average.

$$SE = s / \sqrt{n}$$

$$95\% \text{ interval} = \text{average} \pm 1.96 \times SE$$

For a continuous metric, s is the standard deviation of the individual values and n is how many of them there are. A rate gives you its own standard deviation, so p and n are enough; an average does not, so you need the values themselves. In Excel that is =STDEV.S(range)/SQRT(COUNT(range)).

A CONFIDENCE INTERVAL ON SAQUON BARKLEY'S YARDS PER CARRY

436 carries, averaging 5.74 yards, with a standard deviation of 10.47

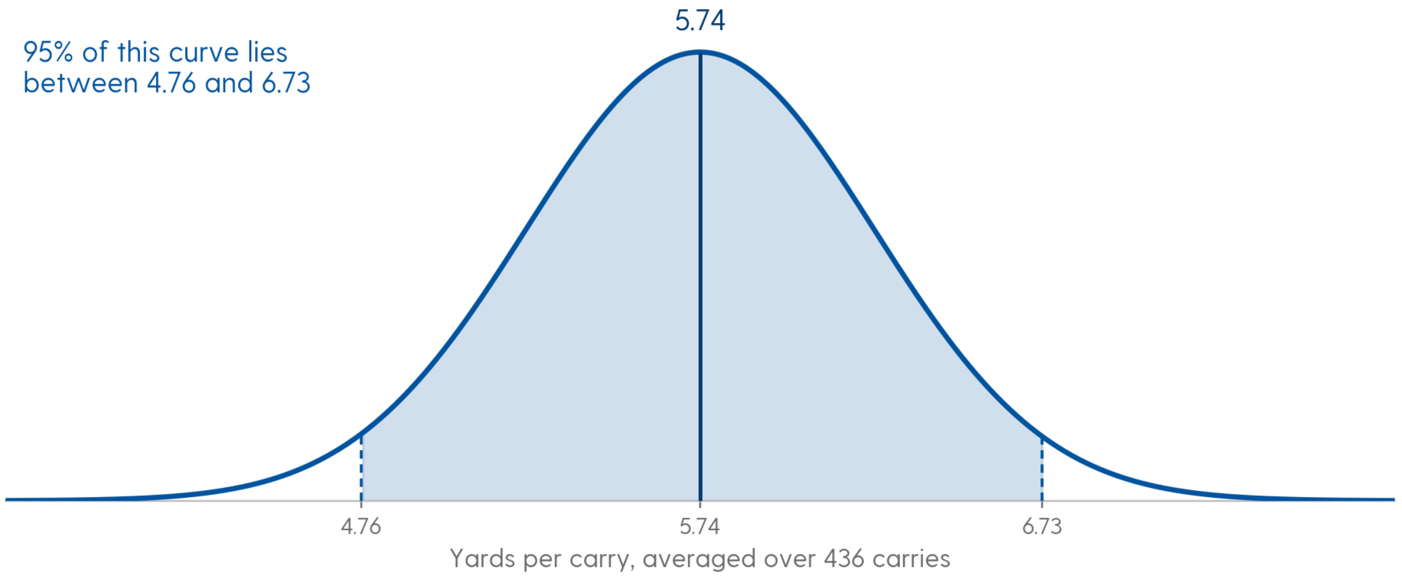
standard error = 10.47 / square root of 436 = 0.502

95% interval = 5.74 plus or minus 1.96 x 0.502 = 4.76 to 6.73

For a continuous metric's average, we will calculate the standard error. This is the standard deviation divided by the square root of n. We will then look at 1.96 standard errors to each side of the mean, which people round off to two, to determine a 95% confidence interval.

SOMEWHERE BETWEEN 4.76 AND 6.73 YARDS A CARRY

WHERE SAQUON BARKLEY'S TRUE YARDS PER CARRY COULD SIT



The curve holds the values his average could have taken if he ran the same 436 carries again. Its width is the standard error, 0.50 yards; the individual runs scatter with a standard deviation of 10.47. The shaded middle is 1.97 yards wide, and that is the range his true rate per carry is consistent with.

nflverse play-by-play, 2024 regular season and playoffs

A RATE CARRIES ITS OWN STANDARD ERROR. AN AVERAGE DOES NOT

To get that 0.502 you needed all 436 carries, because the standard deviation comes from the individual runs. A rate needs only two numbers. Gilgeous-Alexander made 601 of 669 free throws, and that pair alone gives you his standard error, his interval, and the chance he makes his next two.

BEFORE YOU PUSH ON TUESDAY, SEPTEMBER 8

Commit the filled-in worksheet in weeks/week02/data/.

Commit your analysis code and its outputs in weeks/week02/outputs/.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Tuesday, September 8 at 11:59pm. Labor Day moves this week's deadline off Monday.